

95% probability
the difference is
in this interval



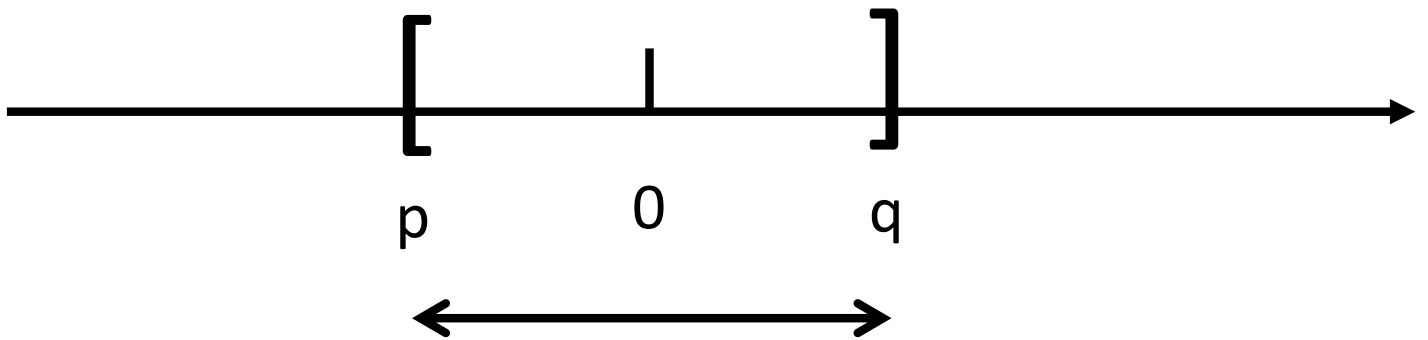
95% probability
the difference is
positive



95% probability
 $\text{EMRT System 2} - \text{EMRT System 1} > 0$



95% probability
 $\text{EMRT System 2} > \text{EMRT System 1}$



95% probability
the difference is
in this interval



95% probability
the difference is
positive or negative

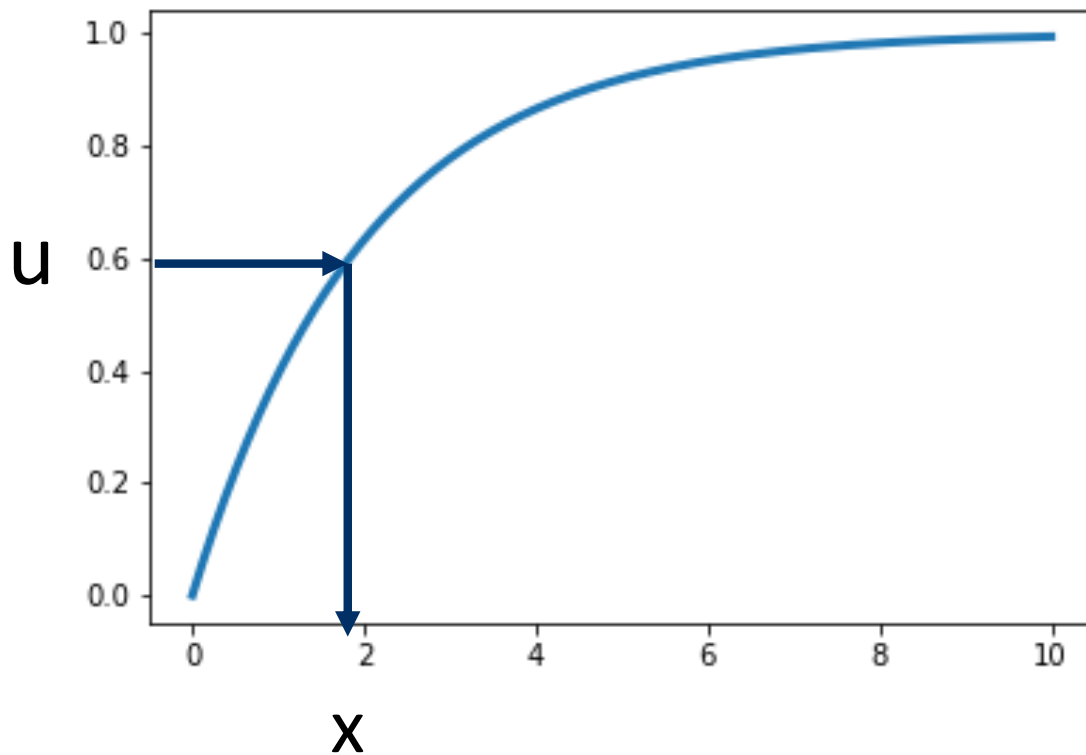
$$Z_1 = 1$$

$$\begin{aligned} Z_2 &= a Z_1 + c \pmod{m} \\ &= 3 * 1 + 0 \pmod{5} \\ &= 3 \end{aligned}$$

$$\begin{aligned} Z_3 &= a Z_2 + c \pmod{m} \\ &= 3 * 3 + 0 \pmod{5} \\ &= 9 \pmod{5} \\ &= 4 \end{aligned}$$

$$\begin{aligned} Z_4 &= 3 * 4 \pmod{5} \\ &= 2 \end{aligned}$$

$$\begin{aligned} Z_5 &= 3 * 2 \pmod{5} \\ &= 1 \end{aligned}$$



$$F(x) = 1 - \exp(-\lambda x)$$

Compute $F^{-1}(u)$

$$x = F^{-1}(u)$$

$$F(x) = u$$

$$1 - \exp(-\lambda x) = u$$

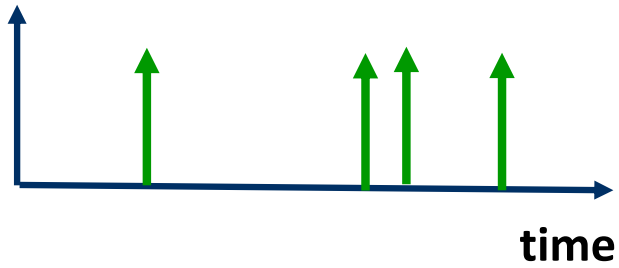
$$\exp(-\lambda x) = 1 - u$$

$$-\lambda x = \log(1 - u)$$

$$x = -\log(1 - u) / \lambda$$

Simulation experiment

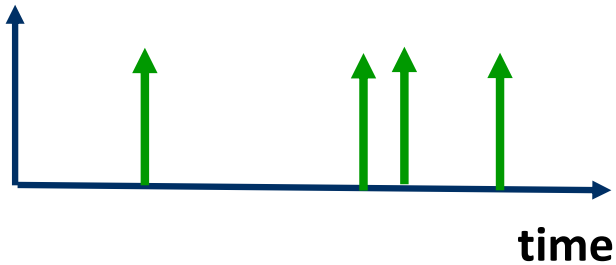
Arrival pattern



Response times of the job
= $X(1)$, $X(2)$, $X(3)$, $X(4)$

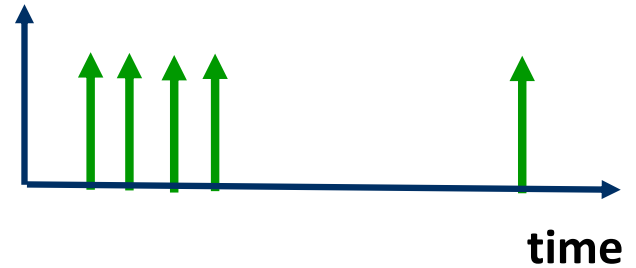
Simulation experiment 1 = Replication 1

Arrival pattern



Response times of the job
= $X(1), X(2), X(3), X(4)$

Simulation experiment 2 = Replication 2



Response times of the job
= $X(1), X(2), X(3), X(4), X(5)$

Same symbol but highly likely
different values

Mean response time of
replication 1
= $T(1)$

Mean response time of
replication 2
= $T(2)$

Note: $t_{[p],n} = t_{n,p}$ in the lecture notes

STATISTICAL TABLES

$$p = 1 - \frac{\alpha}{2}$$

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A.4 QUANTILES OF THE t DISTRIBUTION

Table A.4 lists $t_{[p],n}$. For example, the $t_{[0.95],13}$ required for a two-sided 90% confidence interval of the mean of a sample of 14 observation is 1.771.



TABLE A.4 Quantiles of the t Distribution

n	p							
	0.6000	0.7000	0.8000	0.9000	0.9500	0.9750	0.9950	0.9995
1	0.325	0.727	1.377	3.078	6.314	12.706	63.657	636.619
2	0.289	0.617	1.061	1.886	2.920	4.303	9.925	31.599
3	0.277	0.584	0.978	1.638	2.353	3.182	5.841	12.924
4	0.271	0.569	0.941	1.533	2.132	2.776	4.604	8.610
5	0.267	0.559	0.920	1.476	2.015	2.571	4.032	6.869
6	0.265	0.553	0.906	1.440	1.943	2.447	3.707	5.959
7	0.263	0.549	0.896	1.415	1.895	2.365	3.499	5.408
8	0.262	0.546	0.889	1.397	1.860	2.306	3.355	5.041
9	0.261	0.543	0.883	1.383	1.833	2.262	3.250	4.781
10	0.260	0.542	0.879	1.372	1.812	2.228	3.169	4.587
11	0.260	0.540	0.876	1.363	1.796	2.201	3.106	4.437
12	0.259	0.539	0.873	1.356	1.782	2.179	3.055	4.318
13	0.259	0.538	0.870	1.350	1.771	2.160	3.012	4.221
14	0.258	0.537	0.868	1.345	1.761	2.145	2.977	4.140
15	0.258	0.536	0.866	1.341	1.753	2.131	2.947	4.073
16	0.258	0.535	0.865	1.337	1.746	2.120	2.921	4.015
17	0.257	0.534	0.863	1.333	1.740	2.110	2.898	3.965
18	0.257	0.534	0.862	1.330	1.734	2.101	2.878	3.922
19	0.257	0.533	0.861	1.328	1.729	2.093	2.861	3.883
20	0.257	0.533	0.860	1.325	1.725	2.086	2.845	3.850
21	0.257	0.532	0.859	1.323	1.721	2.080	2.831	3.819
22	0.256	0.532	0.858	1.321	1.717	2.074	2.819	3.792
23	0.256	0.532	0.858	1.319	1.714	2.069	2.807	3.768
24	0.256	0.531	0.857	1.318	1.711	2.064	2.797	3.745
25	0.256	0.531	0.856	1.316	1.708	2.060	2.787	3.725
26	0.256	0.531	0.856	1.315	1.706	2.056	2.779	3.707
27	0.256	0.531	0.855	1.314	1.703	2.052	2.771	3.690
28	0.256	0.530	0.855	1.313	1.701	2.048	2.763	3.674
29	0.256	0.530	0.854	1.311	1.699	2.045	2.756	3.659
30	0.256	0.530	0.854	1.310	1.697	2.042	2.750	3.646
60	0.254	0.527	0.848	1.296	1.671	2.000	2.660	3.460
90	0.254	0.526	0.846	1.291	1.662	1.987	2.632	3.402
120	0.254	0.526	0.845	1.289	1.658	1.980	2.617	3.375